



BI Agentic Platform

AI-Powered Business Intelligence & Forecasting System

Transforming Voice, Data, and AI into Intelligent Business Decisions

Superviosers: Anas Abdualaziz

Student: Massa Nasri , Ayman Al Kotyfan

Problems & Motivation

Transforming Business Intelligence into an AI-driven conversational experience.



SQL Dependency

Traditional BI requires SQL expertise — blocking non-technical users.



Static Dashboards

Existing dashboards lack interactivity and real-time adaptability.



Delayed Decisions

Manual analysis slows decision-making at every level.



Complex Forecasting

Forecasting tools demand ML expertise most teams lack.

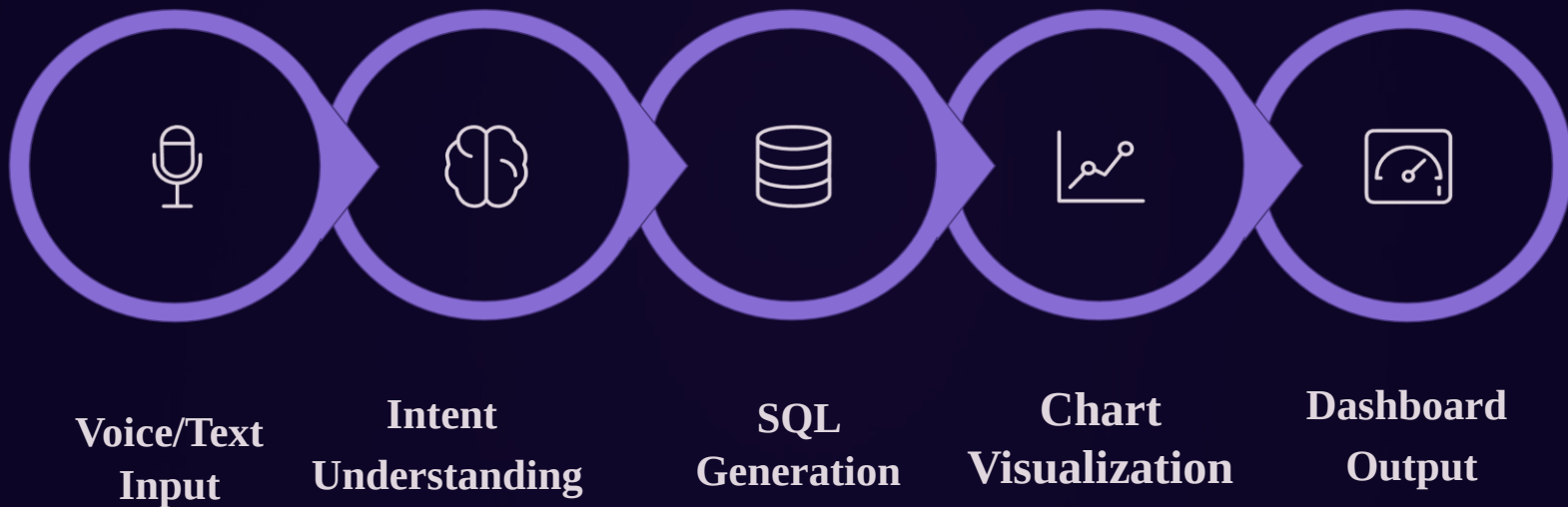


Voice-Driven Insights

Companies need faster, voice-powered analytical access.

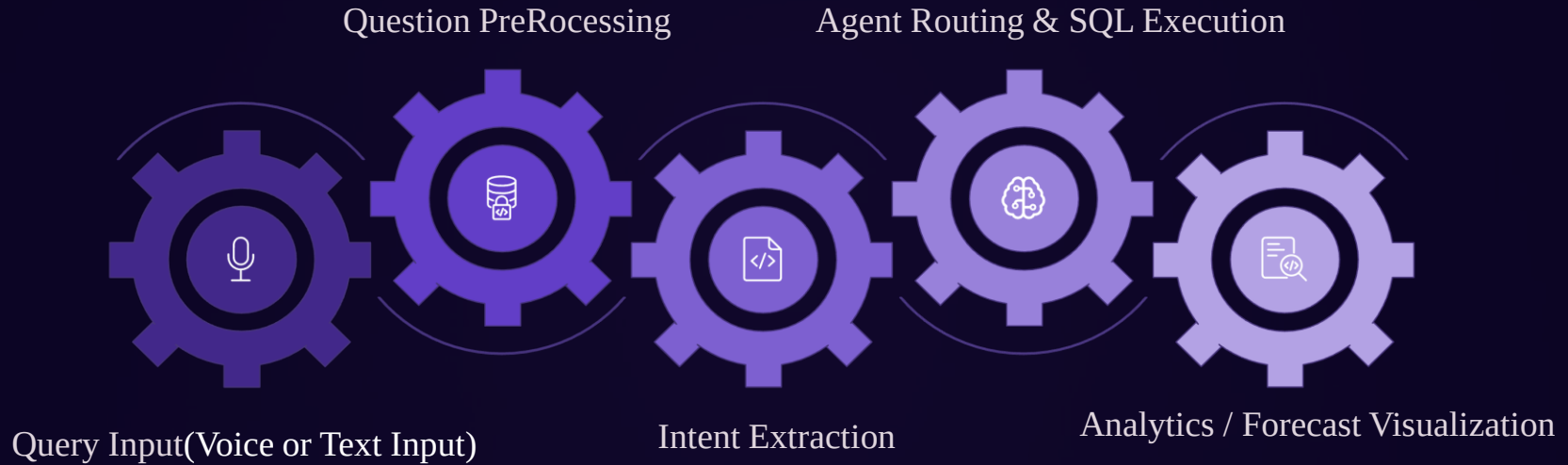
What is BI Voice Agent?

An AI-powered business intelligence system that lets users ask analytical questions via voice — automatically generating SQL queries, executing them on ClickHouse, and visualizing results through interactive dashboards.



From natural language question to actionable insight — fully automated, zero SQL required.

BI Agentic Platform Overview?



The system processes user queries through intelligent AI and data layers to deliver advanced analytics and enterprise services.

01

Software Engineering



System Actors

Four distinct user roles, each with tailored access levels and capabilities within the BI Agentic Platform.



Manager

Asks analytical questions, reviews KPIs, and monitors team performance through voice or text queries.



Analyst

Builds dashboards, configures data sources, and generates advanced reports and forecasts.



Executive

Consumes high-level insights and strategic summaries for fast, informed decision-making.



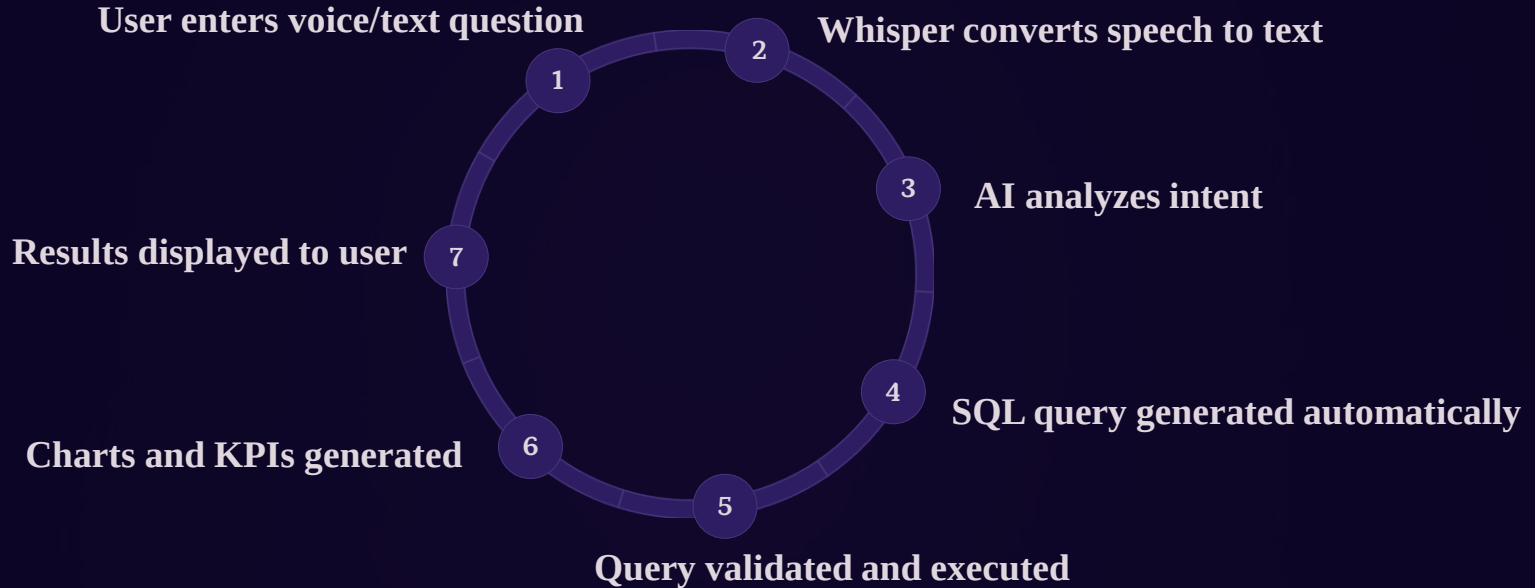
System Admin

Manages users, roles, integrations, and platform-wide configuration and security.

Functional Requirement — Ask Question

R-16

ACTOR: MANAGER



Alternative Flows

- Unsupported query → user notified
- Voice recognition failure → retry prompt
- SQL execution failure → error logged

Postconditions

- Analytical
- Visualization
- System

Non-Functional Requirements



Performance

Fast response time and efficient query execution.



Scalability

Supports growing users and expanding datasets.



Security

Secure authentication and protected data access.



Reliability

Stable operation with minimal downtime.



Usability

Simple and intuitive interfaces for all users.



Maintainability

Easy upgrades and modular maintenance.



Compatibility

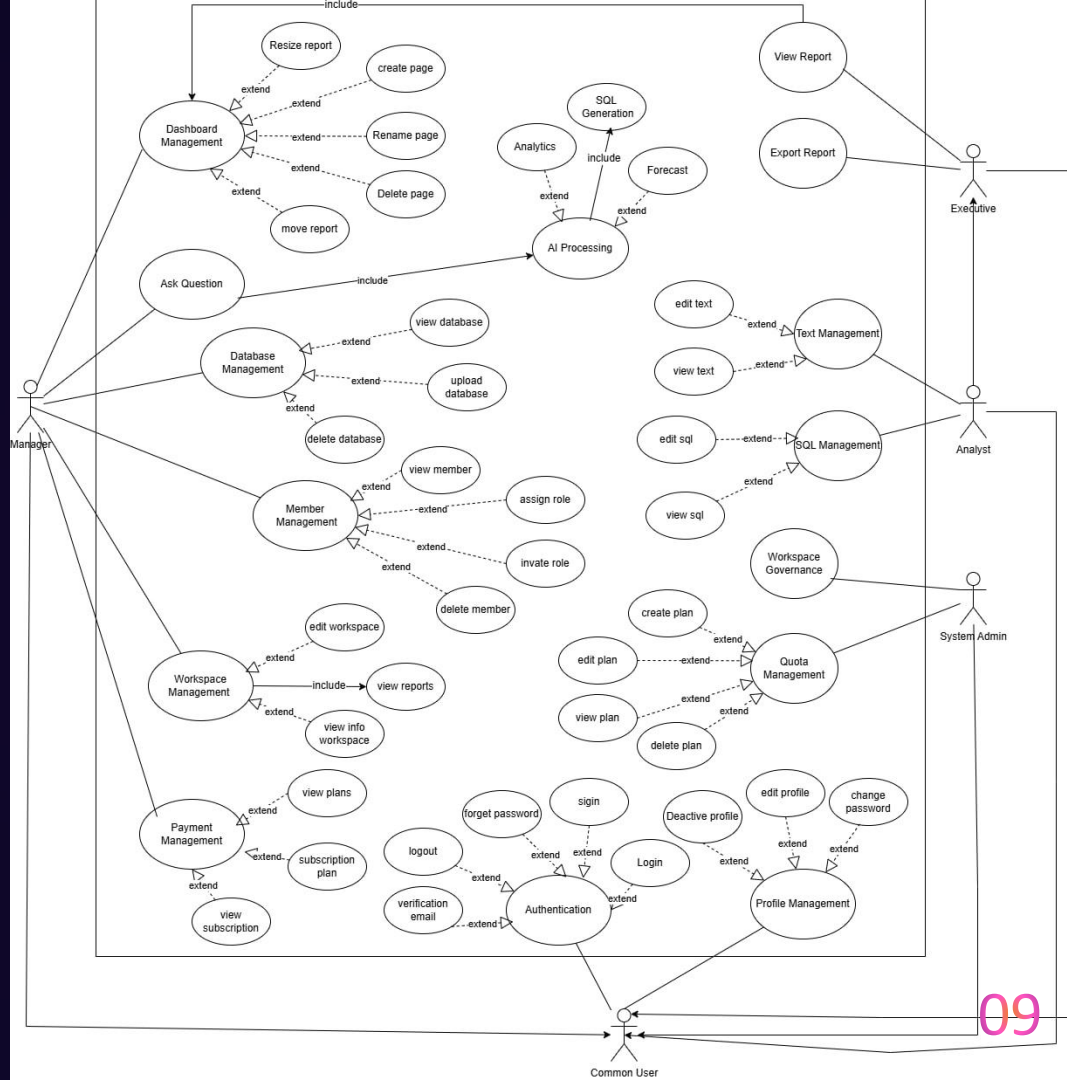
External tool and database compatibility.



Portability

Deployable across environments via containers.

Use Case Diagram

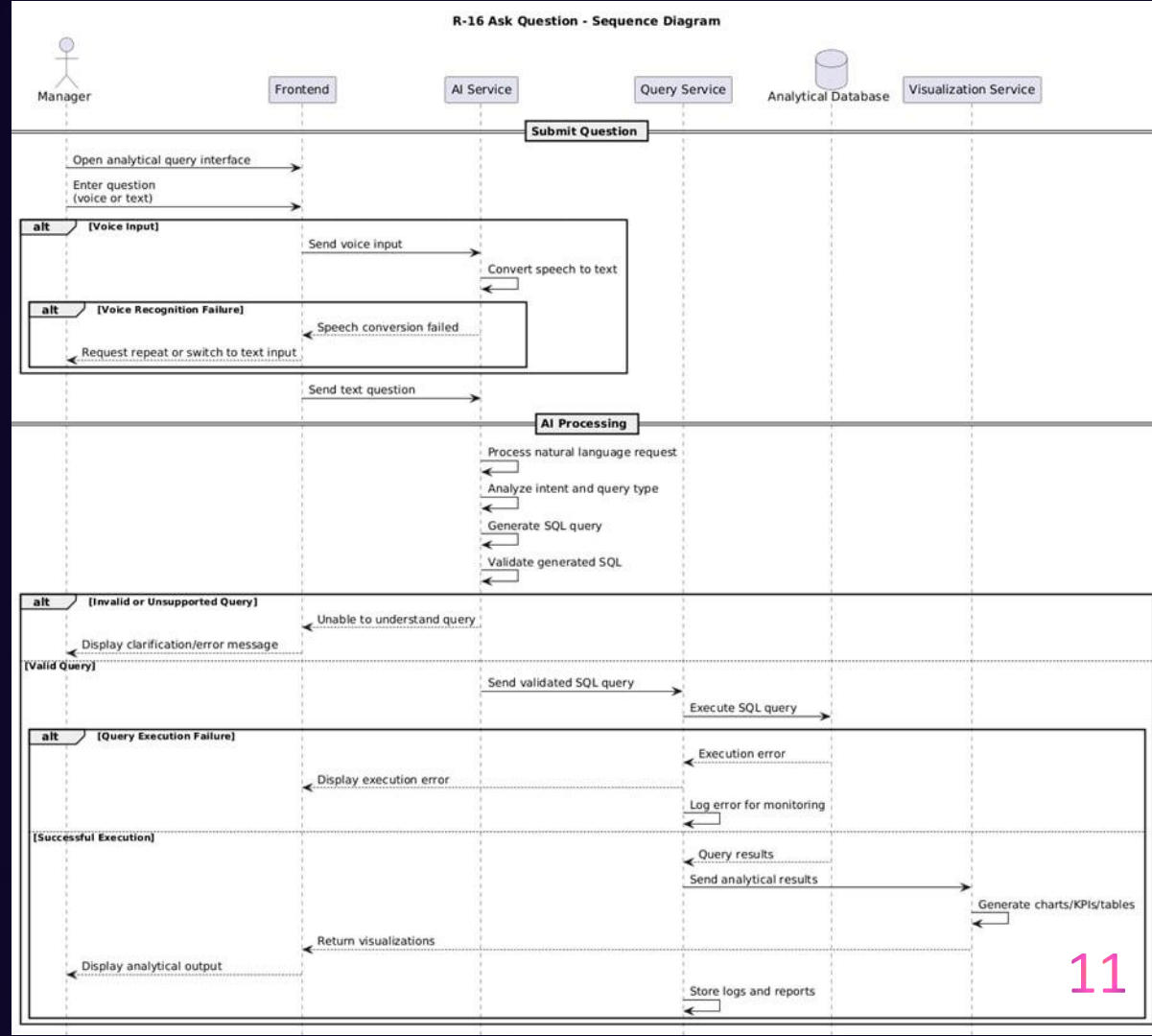


Requirement Table (Ask question)

Requirement ID	R-16
Requirement Name	Ask Question
Actors	Manager
Preconditions	1. The user must be authenticated and logged into the system. 2. The user must have access to a workspace with connected datasets or analytical databases. 3. Voice input requires microphone access permission.
Main Flow	1. The user opens the analytical query interface. 2. The user enters a question using either voice or text input. 3. If voice input is used, the system converts speech into text. 4. The system processes the natural language request. 5. The system analyzes the user intent and determines the query type. 6. The system generates and validates the appropriate SQL query automatically. 7. The system executes the query on the analytical database. 8. The system generates charts, KPIs, tables, or analytical visualizations based on the query result. 9. The system displays the generated analytical output to the user.

Alternative Flows	A1 - Invalid or Unsupported Query 1. The system cannot understand or classify the user request. 2. The system displays an error or clarification message requesting a more specific query. A2 - Voice Recognition Failure 1. The system fails to convert speech into text correctly. 2. The system requests the user to repeat the voice input or switch to text input. A3 - Query Execution Failure 1. The generated SQL query fails during execution. 2. The system displays an execution error message. 3. The system logs the error for monitoring and debugging purposes.
Postconditions	1. The user receives analytical results based on the submitted question. 2. The generated report or visualization is displayed successfully. 3. The query and generated outputs may be stored for dashboard usage or future access. 4. System logs are updated for monitoring and auditing purposes.

Sequence Diagram (Ask quastion)

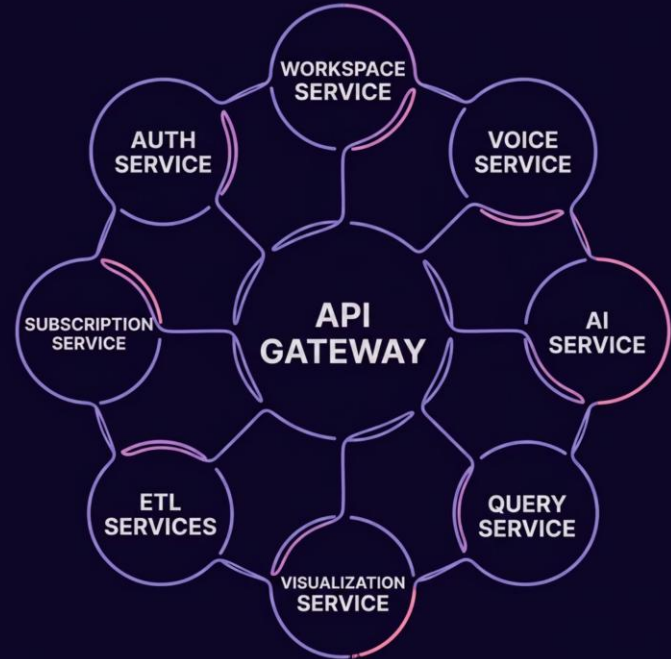


Microservices Architecture

Modular, scalable, and independently deployable services — each designed to operate, scale, and update without impacting the broader system.

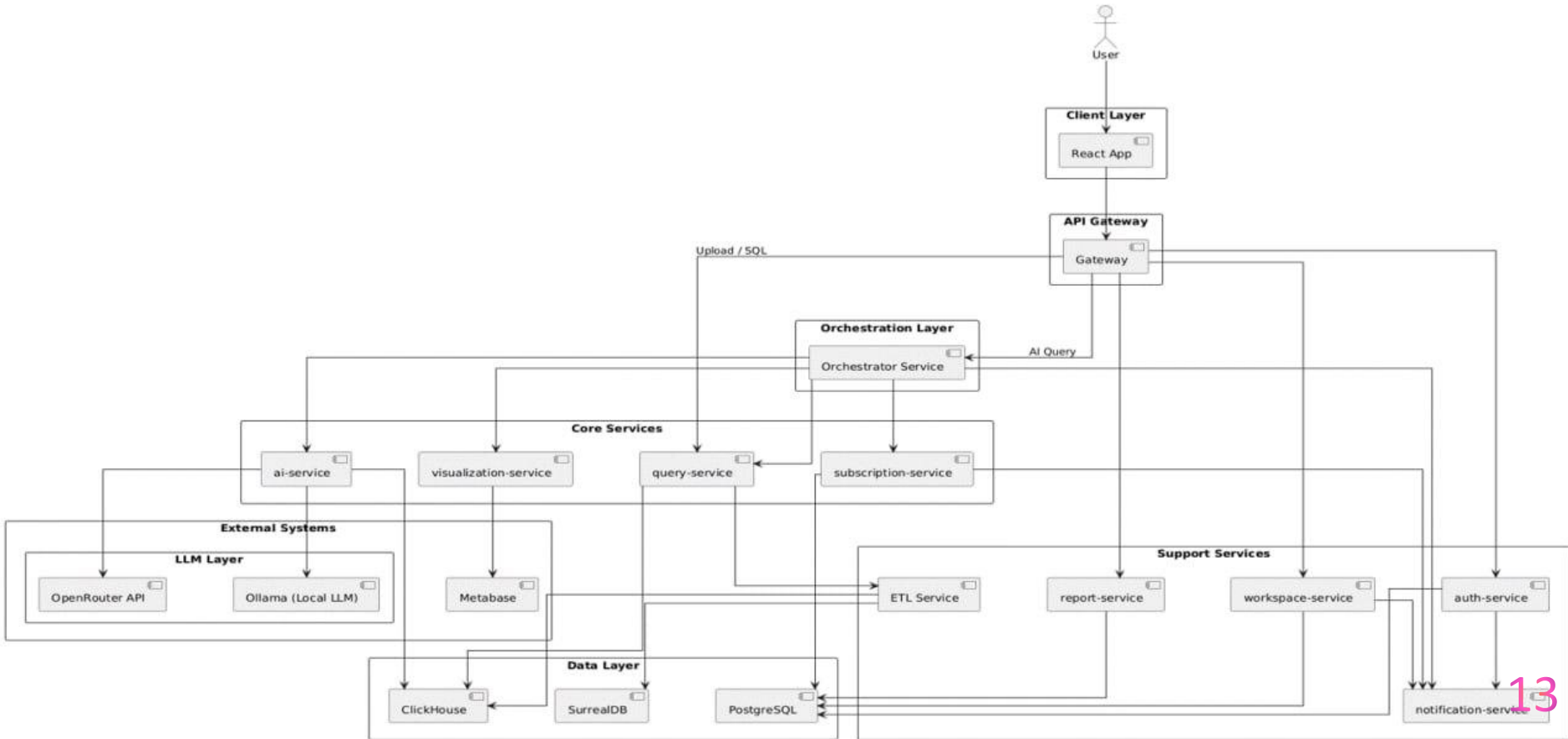
- **API Gateway** — Central routing layer
- **Auth Service** — Authentication & access control
- **Voice Service** — Speech processing via Whisper
- **AI Service** — Intent classification & SQL generation
- **Query Service** — ClickHouse execution engine
- **Visualization Service** — Metabase dashboard rendering
- **ETL Services** — Data pipeline orchestration
- **Subscription Service** — Plans & billing management

Core Services

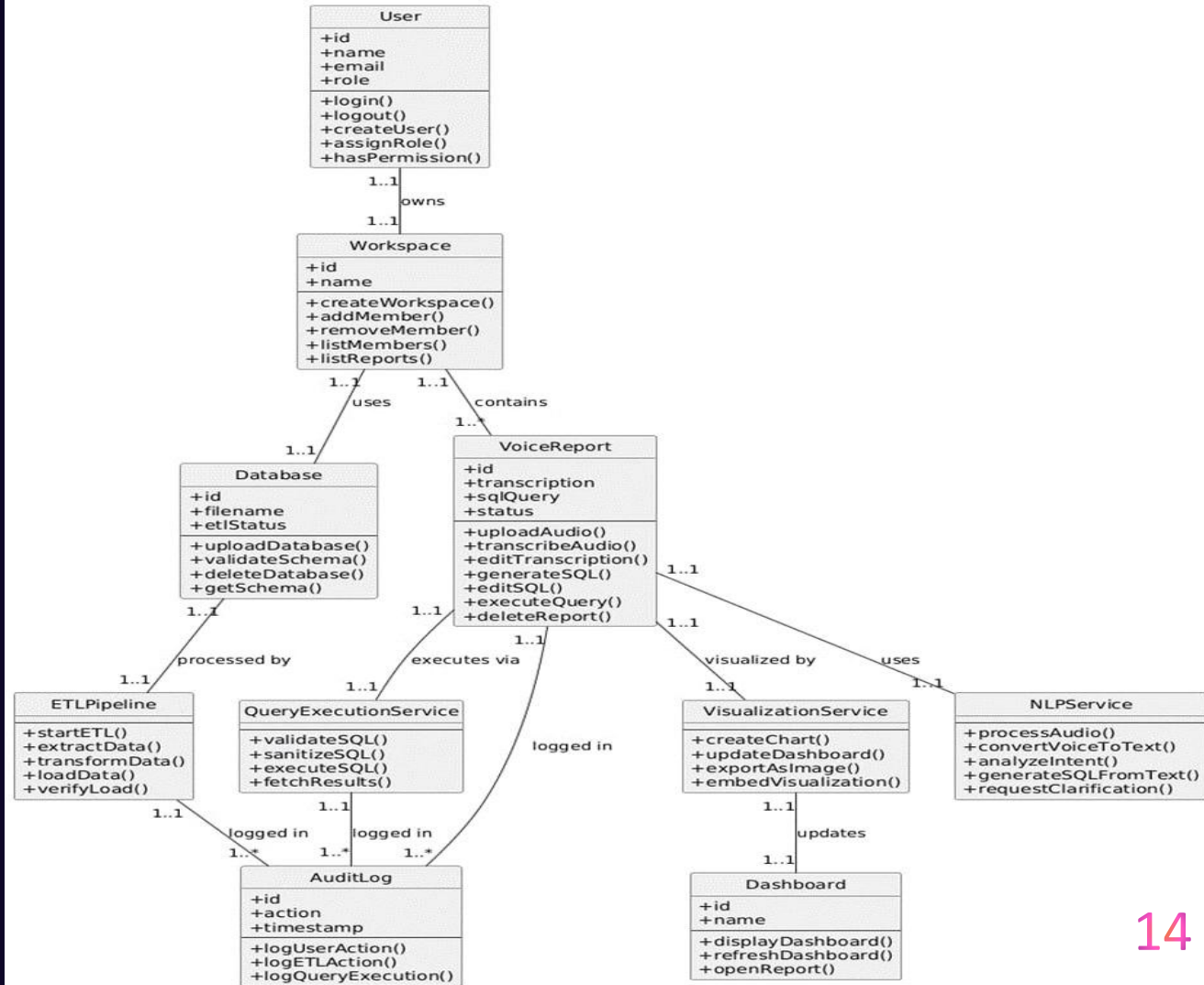


System Architecture

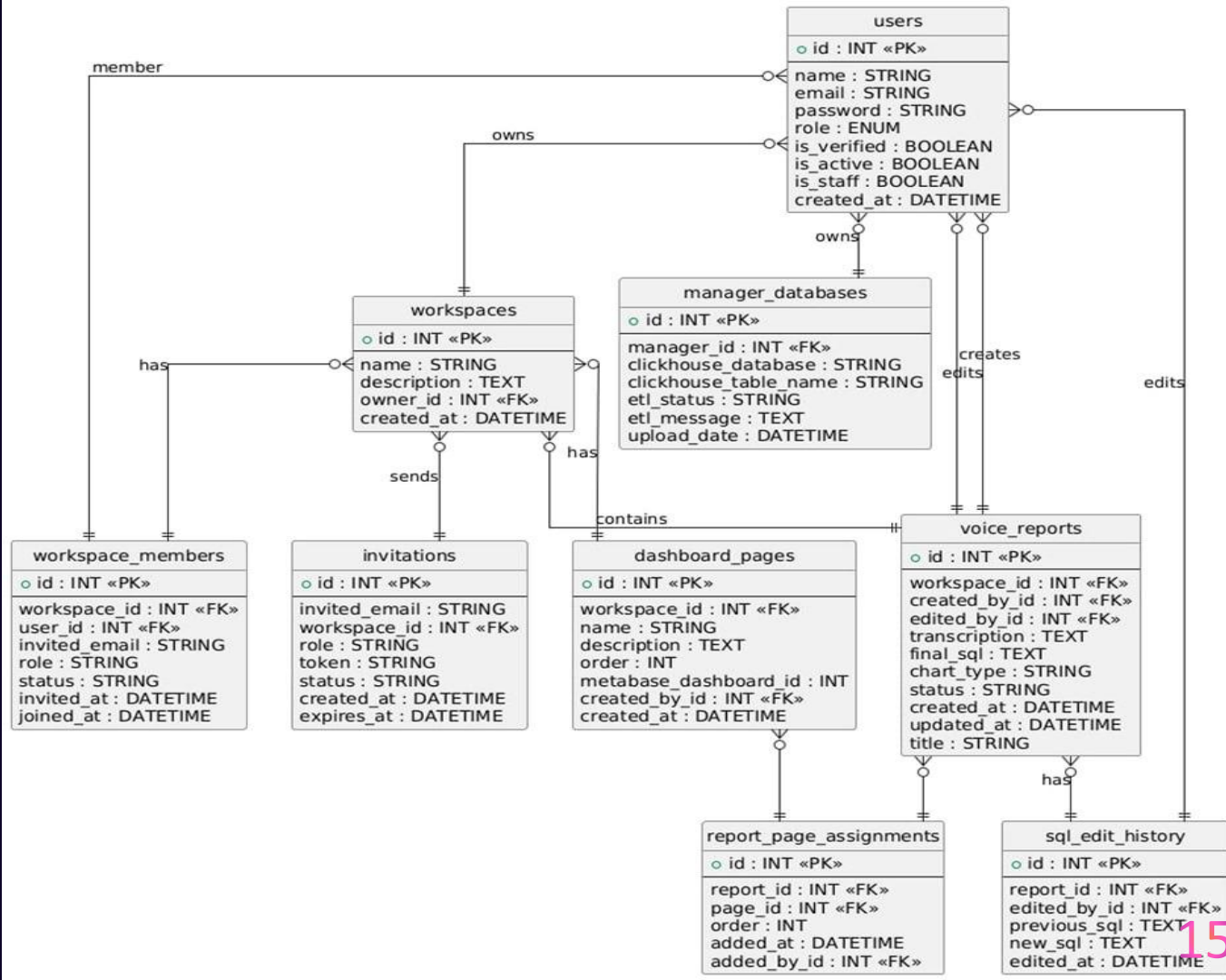
BI-Agentic Platform



Class Diagram

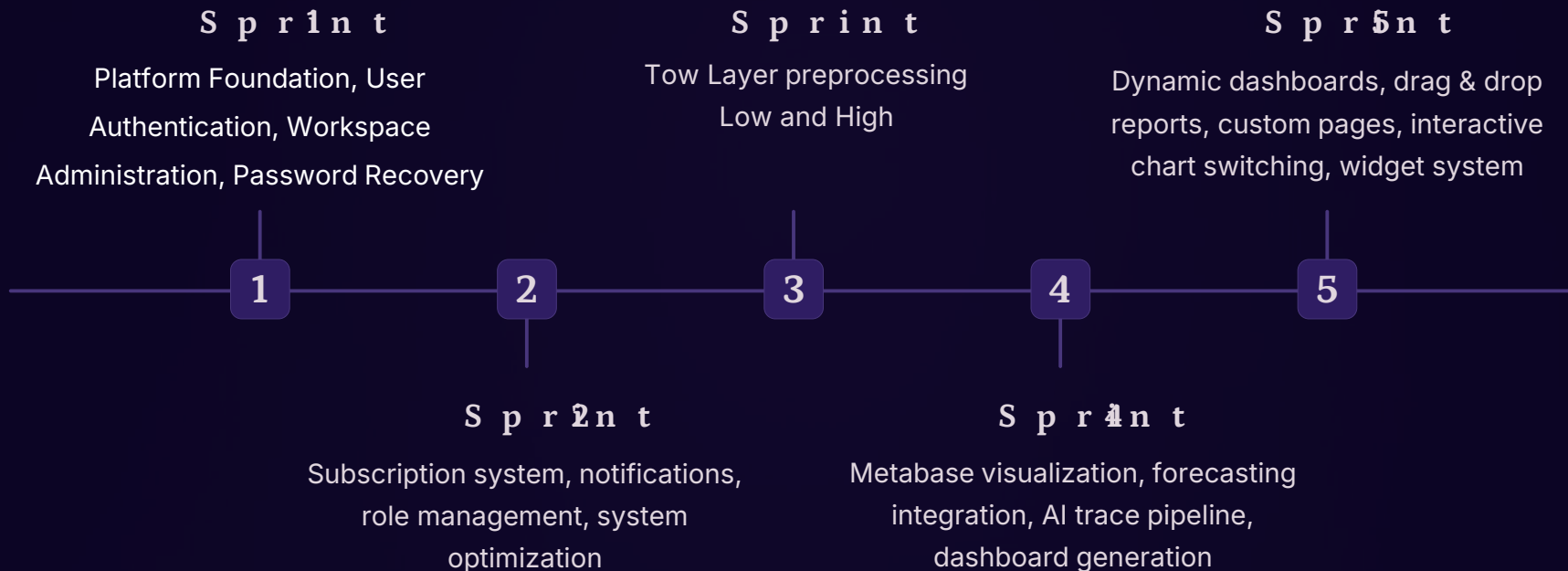


ERD Diagram



Project Sprints

Five focused sprints delivering a production-ready AI analytics platform — from core architecture to advanced interactive dashboards.



02 Artificial Intelligence



Reference Study Comparison Table for Time Series Forecasting Models

Model	Type	Requires Training	API Availability	Deployment	Pricing	Key Strengths	Key Limitations
TimeGPT	Time-Series Foundation Model (Transformer-based)	No (Zero-shot supported)	Yes (API + Python SDK)	Cloud (Nixtla API, optional Azure integration)	Paid (after free trial)	High accuracy, zero-shot forecasting, easy API use, supports multivariate & exogenous variables	Not open-source, requires internet, paid, specialized for time series only
DeepAR+	Probabilistic Time-Series Model (RNN-based)	Yes	Indirect (via AWS SageMaker / legacy AWS Forecast)	Cloud (AWS)	Paid	Learns global patterns across multiple time series, probabilistic forecasting, handles seasonality and covariates	Requires large data, training overhead, AWS dependency (for managed version), not zero-shot
BigQuery ML	Statistical Time-Series Model	Yes (automatic model fitting)	Yes (SQL + API)	Cloud (Google BigQuery)	Paid (pay-as-you-go)	SQL-based usage, automatic model fitting, scalable, built-in anomaly detection, easy integration with data warehouse	Limited flexibility compared to deep learning models, requires cloud usage, less suitable for complex patterns
TimesFM	Time-Series Foundation Model (Transformer-based)	No (Zero-shot supported)	Yes (Python API / local inference)	Local or Cloud (flexible)	Free (open-source)	No training required, fast inference, strong zero-shot forecasting, supports long sequences, provides prediction intervals	Limited customization, requires correct input formatting, may underperform on domain-specific data
StatsForecast	Statistical Time-Series Library	Yes (model fitting)	No official API (Python library)	Local	Free (open-source)	Very fast, scalable, strong statistical baselines	Limited in capturing complex patterns, requires coding
Prophet	Statistical Time-Series Model	Yes (automatic model fitting)	No official API (Python/R library)	Local	Free (open-source)	Easy to use, interpretable, handles seasonality and missing data	Less effective for complex patterns, limited flexibility
Chronos	Time-Series Foundation Model (Transformer-based)	No (Zero-shot supported)	No official API (via Hugging Face / SageMaker)	Local or Cloud	Free (open-source) / Paid (cloud usage)	Zero-shot forecasting, transformer-based, supports fine-tuning, state-of-the-art performance	Requires setup, computational resources, no direct API

Comparative Analysis of Similar Systems and the Target System

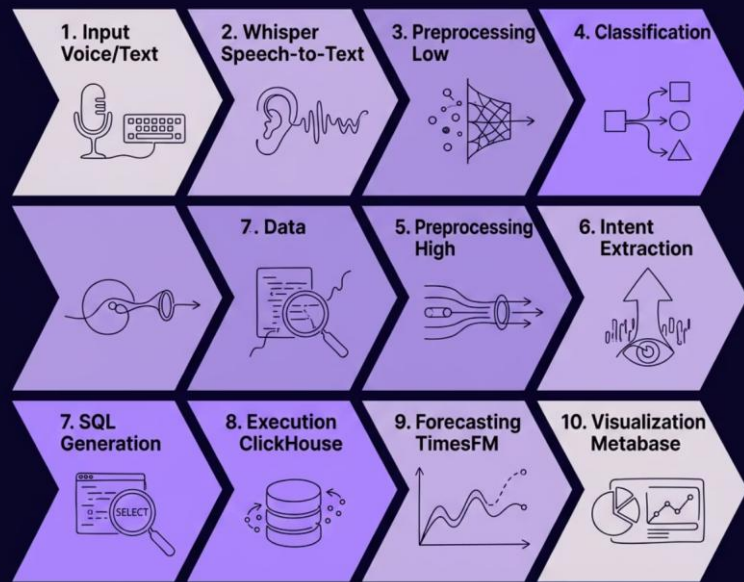
الميزة	BI Voice Agent	Zoho Zia	QuickSight Q	Power BI Copilot	Looker Gemini	Databricks Genie	Tableau AI	Tellius	ThoughtSpot	Cognos	Oracle
Agentic Reasoning	☑	◇	◇	✗	✗	☑	◇	◇	✗	✗	✗
Multi-step Analysis	☑	◇	◇	✗	✗	☑	◇	◇	✗	✗	✗
Natural Language Query	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑
Auto Visualization	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑
Voice Query	☑	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
Question Classification	☑	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
Router Agent	☑	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
Streaming ETL Pipeline	☑	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
Custom Analytics Engine	ClickHouse	Zoho Engine	AWS SPICE	Fabric	BigQuery	Lakehouse	Hyper	External DB	External DB	IBM Engine	Oracle DB
Forecasting	☑	☑	☑	◇	◇	☑	◇	☑	◇	☑	☑
Anomaly Detection	☑	☑	☑	◇	◇	☑	◇	☑	◇	◇	☑
Multi-Tenant SaaS	☑	◇	✗	✗	✗	✗	✗	✗	✗	✗	✗

End-to-End AI Pipeline

Fully automated analytical reasoning pipeline — from raw voice input to intelligent visualized insights, with zero manual intervention.

Pipeline Highlights

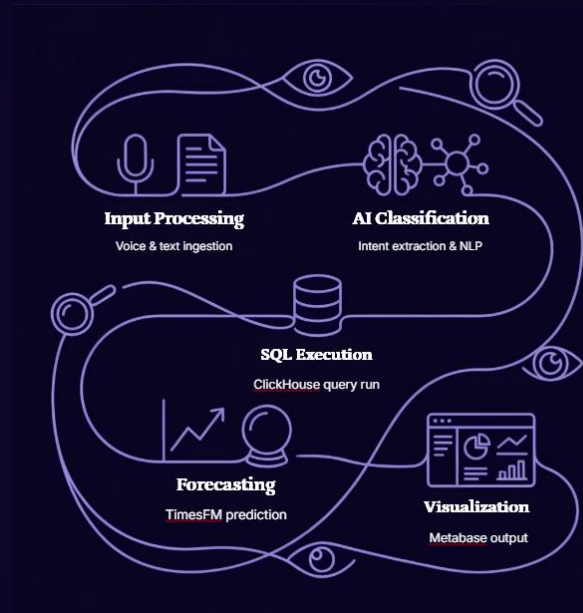
- **Whisper** — Converts voice to accurate text
- **Intent Extraction** — AI classifies user intent
- **SQL Generation** — Automatic query construction
- **ClickHouse** — High-speed query execution
- **TimesFM** — Zero-shot time-series forecasting
- **Metabase** — Interactive result visualization



Dagster Pipeline Orchestration

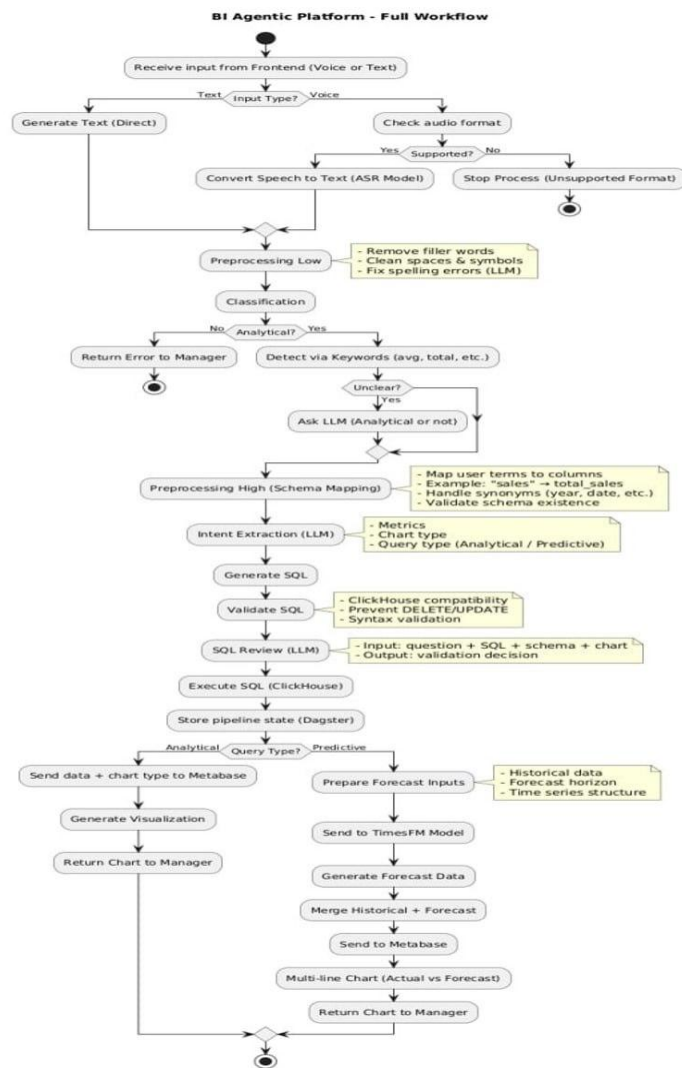
Dagster orchestrates and monitors every AI pipeline stage—ensuring reliability, traceability, and scalability across the platform.

- Asset-based pipeline orchestration
- Stage monitoring and tracing
- Modular AI workflow execution
- Retry and failure handling
- Pipeline observability
- Scalable orchestration architecture



✓ Dagster enables reliable and maintainable AI workflow automation.

AI Pipeline



TimesFM Forecasting

TimesFM — Zero-Shot Forecasting

Google's foundation model for time-series forecasting — accurate predictions without retraining on company-specific datasets.

- Zero-shot forecasting capability
- Seamless SQL integration
- Historical + predictive visualization
- No retraining required



03 Data Science



Two Layers of Preprocessing

Low-Level Preprocessing

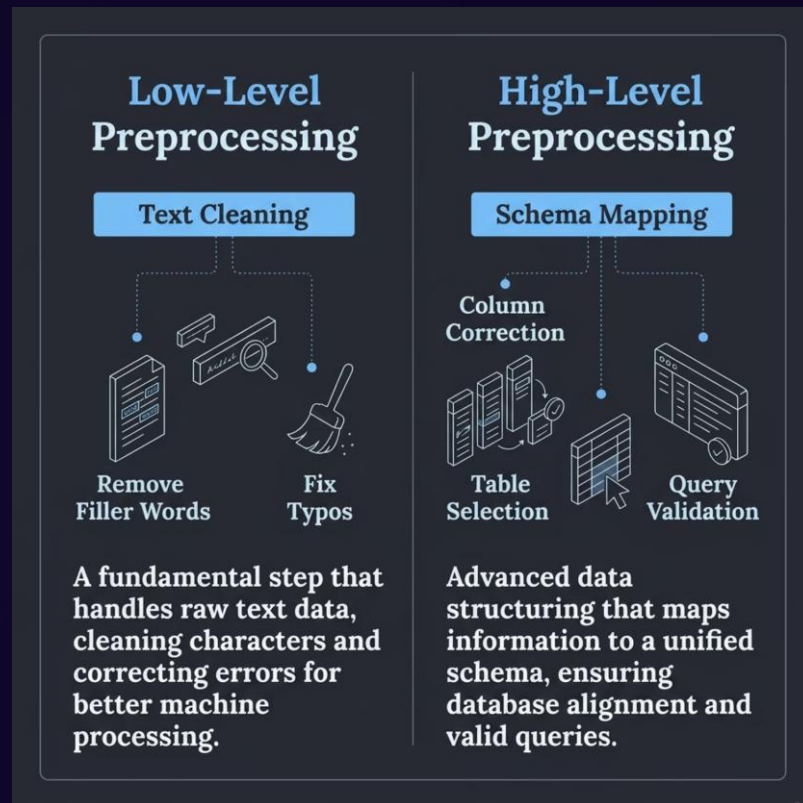
Cleans raw input before any AI processing:

- Text normalization & cleaning
- Removal of filler words and noise
- Automatic typo correction

High-Level Preprocessing

Bridges natural language to database structure:

- Schema and entity mapping
- Column name correction & resolution
- Table selection & query validation



Technology Stack



Backend

Python · Django



AI Models

Whisper · OpenRouter · Ollama · TimesFM



Infrastructure

Docker · Kafka · Git · Metabase · Dagster



Frontend

React · Axios · Tailwind

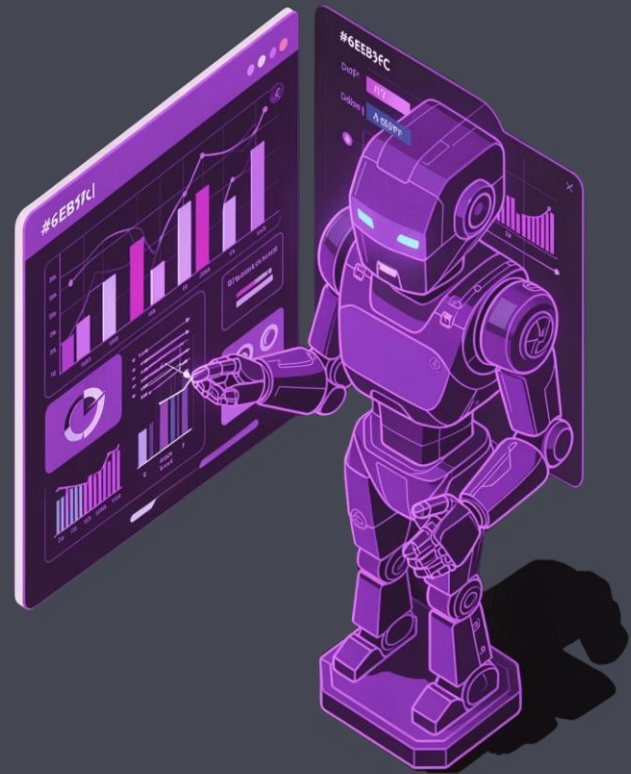


Databases

PostgreSQL · ClickHouse · SurrealDB



04 Our System



Sprint 1

Platform Foundation, User Authentication, Workspace Administration, Password Recovery

SYSTEM ADMIN

Admin Dashboard

Manage plans, monitor platform usage, and control subscription lifecycle.

+ Create Plan

[→ Logout]

Users
3

Workspaces
2

Active Subscriptions
1

Revenue
\$58.00

Subscription Plans

Create, edit, activate, and retire plans without disrupting existing subscriptions.

ayman

Active

popular

sfzfd

montly

Active

asd

asd



Forgot Password

Enter your email to receive a reset code.

Email Address

you@example.com

Send Verification Code

[Back to Sign In](#)

Sprint 2

Subscription system, notifications, role management, system optimization

New Analytical Report

Hello **massa nasri**,

A new analytical report #335 was created in **Ayman Alkotyfan's Workspace** by aymannk331@gmail.com.

BI Voice Agent Notifications

asd

montly

asd

\$29.00 / MONTH

\$299.00 yearly

30 days
Duration

100
Voice Requests

⚡ MCP Not Included

- ✓ Priority support
- ✓ Voice analytics
- ✓ Workspace insights

Select Plan

Sprint 3

Tow Layer preprocessing Low and High

4. Preprocessing High success 

SELECTED TABLE
sales_3months_realistic_csv

SELECTED COLUMNS
ds, total_sales, orders, customers

CORRECTIONS

Sprint 4

Metabase visualization, forecasting integration, AI trace pipeline, dashboard generation

AI Trace

1. Original Question success

Compared total sales and customer over time

Re-run

2. Preprocessing Low success

ORIGINAL SENTENCE

Compared total sales and customer over time

CLEANED SENTENCE

Compared total sales and customer over time

CHANGES

No significant changes.

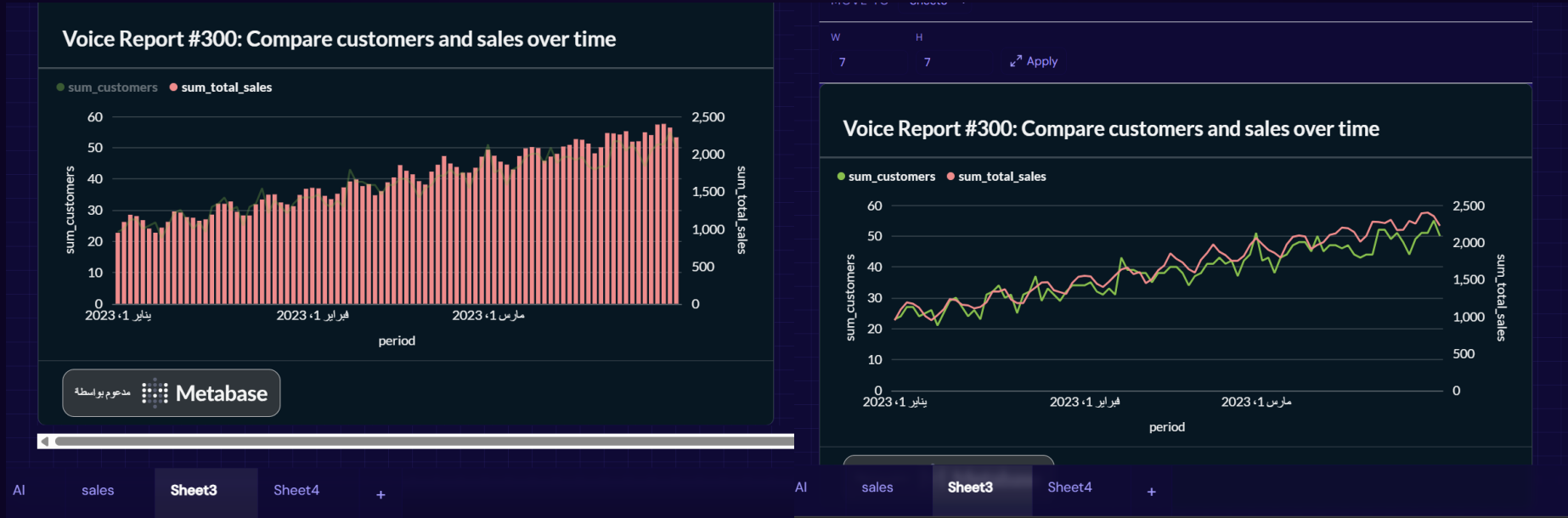
3. Classification success

Voice Report #237: Predict the future relationship between customers



Sprint 5

Dynamic dashboards, drag & drop reports, custom pages, interactive chart switching, widget system



Thank You For Listening!

Lets Watch Demo

